SHEIKH NAZMUL ISLAM

AI/ML Engineer

Mohammadpur, Dhaka, Bangladesh | +8801838457404 | sheikhnazmulislamnir@gmail.com | LinkedIn | GitHub

Professional Summary

AI/ML Engineer with hands-on experience in machine learning, LLM-powered applications, backend API development, and workflow automation. Skilled in Python, FastAPI, LangChain, RAG, vector retrieval, and model deployment. Experienced in building ML prediction systems, document question-answering solutions, and secure REST APIs. Passionate about reliable, scalable AI solutions that solve real-world business problems.

Technical Skills

Programming: Python, SQL, JavaScript, TypeScript
AI/ML: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, AI Agents, Prompt Engineering, Computer Vision
AI Frameworks & Libraries: LangChain, FAISS, Sentence Transformers, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch
LLM Integration: Google Gemini, Groq API, LLM API Integration, Semantic Search, Vector Retrieval, Structured Outputs
Backend & Databases: FastAPI, REST APIs, Webhooks, JSON, Pydantic, SQLAlchemy, PostgreSQL, MySQL, JWT
Automation: n8n, Meta WhatsApp Cloud API, Google Sheets API
Tools & Deployment: Git, GitHub, Docker, Docker Compose, Postman, Streamlit, Gradio, Hugging Face, Render
AI Development Tools: Codex, Claude, Antigravity
Soft Skills: Problem Solving, Analytical Thinking, Critical Analysis, Teamwork, Adaptability, Communication

Project Experience

Plant Disease Detection

LIVE | HuggingFace | GitHub

Python | *TensorFlow* | *EfficientNet-B0* | *Computer Vision*

- Built a chilli leaf disease classifier using EfficientNet-B0 transfer learning in TensorFlow.
- Published the chilli plant disease image dataset on Hugging Face for training and evaluation.
- Deployed a live Hugging Face Spaces app for real-time leaf image prediction.

Diabetes Prediction System

LIVE | GitHub

Python | *XGBoost* | *Scikit-learn* | *Gradio* | *Hugging Face*

- Developed a diabetes prediction application using a trained XGBoost classification model.
- Applied feature scaling with a saved preprocessing artifact before model inference.
- Built an interactive interface to collect eight input features and display predictions.

Expense Tracker API

GitHub | Live

Python | *FastAPI* | *SQLAlchemy* | *PostgreSQL (Neon)* | *Pydantic* | *JWT* | *Passlib* | *Pytest* | *Render*

- Built a RESTful expense API with secure JWT authentication and user-specific transaction access.
- Developed CRUD for income/expense records with filtering by type, category, and amount range.
- Implemented Pydantic validation, SQLAlchemy models, Pytest coverage, and Render deployment.

ML Prediction FastAPI

GitHub

Python | *Machine Learning* | *FastAPI* | *REST APIs* | *Gradio* | *Hugging Face*

- Developed an ML prediction application exposing model inference through a FastAPI REST API.
- Handled structured input and prediction responses for API-based model serving.
- Organized the application to integrate machine learning logic with backend services.

Education

Bachelor of Science in Computer Science and Engineering May 2026
Daffodil International University (DIU), Bangladesh

Achievements & Activities

- **Project Track Student**, Phitron AI/ML Batch 1 – Scored 100% in every exam
- **Campus Ambassador**, Phitron – Organized workshops and professional development
- **Participant**, Brain Station 23 Hackathon 2025 – Intent-based E-commerce Search Engine

Languages

Bengali: Native | **English:** Professional working proficiency